Mini-projet 2 : Contrôle d'Accès et Cryptographie

Partie 2A : Configuration des permissions sur un système Linux

1. Création d'Utilisateurs et Groupes (si nécessaire) :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo adduser alice
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
        Full Name []: Alic
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y

┌──(kali㉿kali)-[~/Downloads]
└─$ sudo adduser bob
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bob
Enter the new value, or press ENTER for the default
        Full Name []: Robert
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
```

Création des utilisateurs.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo addgroup dev

┌──(kali㉿kali)-[~/Downloads]
└─$ sudo usermod -aG dev alice
```

Création du groupe dev et ajout de l'utilisateur alice au groupe dev.

2. Création de Fichiers et Répertoires de Test :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ touch test_file.txt

┌──(kali㉿kali)-[~/Downloads]
└─$ echo "Contenu de test" > test_file.txt

┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l test_file.txt
-rw-rw-r-- 1 kali kali 16 Mar 15 10:59 test_file.txt
```

Création du fichier test_file.txt et affichage des permissions.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l test_dir
total 0
```

Affichage des permissions du dossier test_dir.

4. Modification des Permissions avec chmod :

Exercices

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo setfacl -m u:alice:rx test_file.txt
```

chmod ne permet de spécifier l'utilisateur auquel donner des permissions, j'ai donc installé et utilisé acl pour donner les permissions à l'utilisateur alice.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo chown :dev test_file.txt
┌──(kali㉿kali)-[~/Downloads]
└─$ chmod g+rwx test_file.txt
```

Utilisation de la commande chown pour changer le groupe de test_file.txt puis chmod pour donner les permissions au groupe dev.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo chown :dev test_file.txt
┌──(kali㉿kali)-[~/Downloads]
└─$ chmod g+rwx test_file.txt
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 16
-rw-rw-r--  1 kali kali 1675 Mar 15 06:42 ssh-key-2026-02-08.key
-rw-rw-r--  1 kali kali  399 Mar 15 06:42 ssh-key-2026-02-08.key.pub
drwxrwxr-x  2 kali kali 4096 Mar 15 11:00 test_dir
-rwxrwx---+ 1 kali dev    16 Mar 15 10:59 test_file.txt
```

5. Changement de Propriétaire et de Groupe avec chown et chgrp:

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo chown alice test_file.txt
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 16
-rw-rw-r--  1 kali  kali 1675 Mar 15 06:42 ssh-key-2026-02-08.key
-rw-rw-r--  1 kali  kali  399 Mar 15 06:42 ssh-key-2026-02-08.key.pub
drwxrwxr-x  2 kali  kali 4096 Mar 15 11:00 test_dir
-rwxrwx──+ 1 alice dev    16 Mar 15 10:59 test_file.txt
```

Le propriétaire du fichier est maintenant alice.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo chgrp dev test_dir
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 16
-rw-rw-r--  1 kali  kali 1675 Mar 15 06:42 ssh-key-2026-02-08.key
-rw-rw-r--  1 kali  kali  399 Mar 15 06:42 ssh-key-2026-02-08.key.pub
drwxrwxr-x  2 kali  dev  4096 Mar 15 11:00 test_dir
-rwxrwx──+ 1 alice dev    16 Mar 15 10:59 test_file.txt
```

Le groupe de test_dir est maintenant dev.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo chown alice:dev test_file.txt
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 16
-rw-rw-r--  1 kali  kali 1675 Mar 15 06:42 ssh-key-2026-02-08.key
-rw-rw-r--  1 kali  kali  399 Mar 15 06:42 ssh-key-2026-02-08.key.pub
drwxrwxr-x  2 kali  dev  4096 Mar 15 11:00 test_dir
-rwxrwx──+ 1 alice dev    16 Mar 15 10:59 test_file.txt
```

Le propriétaire du fichier est toujours alice et le groupe est bien dev.

Partie 2B : Mise en place d'un chiffrement basique avec OpenSSL.

1.Préparation des Données :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ echo "Ceci est un texte confidentiel." > secret.txt
```

Préparation du fichier et des données.

2. Chiffrement du Fichier avec OpenSSL:

```
┌──(kali㉿kali)-[~/Downloads]
└─$ openssl enc -aes-256-cbc -e -in secret.txt -out secret.enc -k "mon_mot_de_passe"
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

┌──(kali㉿kali)-[~/Downloads]
└─$ ls -la
total 36
drwxr-xr-x   3 kali  kali 4096 Mar 15 11:48 .
drwx------  17 kali  kali 4096 Mar 15 09:43 ..
-rw-rw-r--   1 kali  kali   64 Mar 15 11:46 chiffre.txt
-rw-rw-r--   1 kali  kali   64 Mar 15 11:51 secret.enc
-rw-rw-r--   1 kali  kali   32 Mar 15 11:38 secret.txt
-rw-rw-r--   1 kali  kali 1675 Mar 15 06:42 ssh-key-2026-02-08.key
-rw-rw-r--   1 kali  kali  399 Mar 15 06:42 ssh-key-2026-02-08.key.pub
drwxrwxr-x   2 kali  dev  4096 Mar 15 11:00 test_dir
-rwxrwx---+  1 alice dev    16 Mar 15 10:59 test_file.txt
```

Le fichier a été correctement créé.

3. Visualisation du Fichier Chiffré :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ cat secret.enc
Salted__◆◆◆◆$◆(◆xG◆&d◆◆+◆◆◆u◆◆@>@}N◆1RmD◆◆◆◆◆◆W6ÎNP=◆ST◆
```

Le fichier apparaît comme crypté.

4. Déchiffrement du Fichier avec OpenSSL:

```
┌──(kali㉿kali)-[~/Downloads]
└─$ openssl enc -aes-256-cbc -d -in secret.enc -out secret_decrypt.txt -k "m
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

┌──(kali㉿kali)-[~/Downloads]
└─$ cat secret_decrypt.txt
Ceci est un texte confidentiel.
```

Le fichier est bien décrypté.

Partie 2C : Création et manipulation de certificats auto-signés.

1.Génération d'une Clé Privée:

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -la
total 44
drwxr-xr-x   3 kali  kali 4096 Mar 15 12:09 .
drwx------  17 kali  kali 4096 Mar 15 09:43 ..
-rw-rw-r--   1 kali  kali   64 Mar 15 11:46 chiffre.txt
-rw-------   1 kali  kali 1708 Mar 15 12:09 cle_privee.pem
```

Clé privée correctement générée.

2. Génération d'une Requête de Signature de Certificat (CSR) :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ openssl req -new -key cle_privee.pem -out requete.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:FR
State or Province Name (full name) [Some-State]:Yvelines
Locality Name (eg, city) []:Montesson
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Kairos
Organizational Unit Name (eg, section) []:0
Common Name (e.g. server FQDN or YOUR name) []:Nael
```

3. Génération du Certificat Auto-Signé :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ openssl x509 -req -in requete.csr -signkey cle_privee.pem -out certifica
t.crt -days 365
Certificate request self-signature ok
subject=C=FR, ST=Yvelines, L=Montesson, O=Kairos, OU=0, CN=Nael, emailAddress
=nael.ouasti@orange.fr
```

4. Visualisation du Certificat :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ openssl x509 -in certificat.crt -text -noout

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            7f:38:55:6a:66:f4:08:b9:7b:71:bf:60:f2:73:a6:b9:b6:e6:05:8d
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=FR, ST=Yvelines, L=Montesson, O=Kairos, OU=0, CN=Nael, emai
lAddress=nael.ouasti@orange.fr
        Validity
            Not Before: Mar 15 16:13:40 2026 GMT
            Not After : Mar 15 16:13:40 2027 GMT
        Subject: C=FR, ST=Yvelines, L=Montesson, O=Kairos, OU=0, CN=Nael, ema
ilAddress=nael.ouasti@orange.fr
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:c0:5f:de:7a:2d:99:2a:53:52:fa:54:3e:7d:0c:
                    dd:5c:18:48:6a:9d:68:f2:e1:88:6b:74:34:67:bb:
                    ea:8c:2e:c8:93:96:59:19:64:57:88:a7:28:a5:6e:
                    80:77:d2:e3:74:51:1c:ba:c9:1c:12:ef:f2:1f:fe:
                    bc:7a:c7:6f:df:cf:d3:1c:76:25:2d:83:aa:b4:fa:
                    33:bc:c6:85:71:8e:5a:77:42:02:a0:61:9f:0b:c4:
                    ca:96:7c:1f:55:0b:d5:41:28:70:b2:a9:3d:50:e1:
                    fd:a9:71:0b:95:2c:80:48:a6:ad:56:c9:06:f6:2f:
                    21:ec:11:f4:94:ab:94:eb:8d:79:7f:a7:3f:65:95:
                    2f:29:81:50:c3:61:9e:e2:32:41:d4:6e:fc:9b:9c:
                    93:60:34:91:dd:4b:2d:f5:89:e3:a3:2b:99:bc:3a:
                    3e:e9:16:33:74:78:8a:ac:e6:68:06:78:0c:b3:96:
                    dc:bc:c0:3d:13:ba:18:3e:35:8e:b9:c9:c5:c0:38:
                    fa:9e:35:ff:b8:5a:b3:79:82:a7:3a:47:c9:9e:6e:
                    7d:d8:da:48:07:9f:4d:3a:5f:05:20:9b:38:2f:bb:
                    cd:8f:1e:8c:7d:5f:dc:24:1d:a9:e7:f5:3c:88:05:
                    db:0d:6a:be:58:b7:5f:5a:74:a1:a1:62:9d:ae:0e:
                    ec:27
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
```

5. Manipulation des Certificats :

```
┌──(kali㉿kali)-[~/Downloads]
└─$ openssl x509 -in certificat.crt -outform DER -out certificat.der

┌──(kali㉿kali)-[~/Downloads]
└─$ ^[[200~openssl x509 -in certificat.crt -outform DER -out certificat.der
zsh: bad pattern: ^[[200~openssl

┌──(kali㉿kali)-[~/Downloads]
└─$ openssl x509 -in certificat.crt -pubkey -noout > cle_publique.pem

┌──(kali㉿kali)-[~/Downloads]
└─$ ls -la
total 60
drwxr-xr-x   3 kali  kali 4096 Mar 15 12:16 .
drwx------  17 kali  kali 4096 Mar 15 09:43 ..
-rw-rw-r--   1 kali  kali 1359 Mar 15 12:13 certificat.crt
-rw-rw-r--   1 kali  kali  961 Mar 15 12:16 certificat.der
-rw-rw-r--   1 kali  kali   64 Mar 15 11:46 chiffre.txt
-rw-------   1 kali  kali 1708 Mar 15 12:09 cle_privee.pem
-rw-rw-r--   1 kali  kali  451 Mar 15 12:16 cle_publique.pem
-rw-rw-r--   1 kali  kali 1082 Mar 15 12:12 requete.csr
-rw-rw-r--   1 kali  kali   32 Mar 15 12:06 secret_decrypt.txt
-rw-rw-r--   1 kali  kali   64 Mar 15 11:51 secret.enc
-rw-rw-r--   1 kali  kali   32 Mar 15 11:38 secret.txt
-rw-rw-r--   1 kali  kali 1675 Mar 15 06:42 ssh-key-2026-02-08.key
-rw-rw-r--   1 kali  kali  399 Mar 15 06:42 ssh-key-2026-02-08.key.pub
drwxrwxr-x   2 kali  dev  4096 Mar 15 11:00 test_dir
-rwxrwx---+  1 alice dev    16 Mar 15 10:59 test_file.txt
```

Conversion de format du certificat et extraction de la clé publique.

Partie 2D : Mise en place d'un VPN (OpenVPN)

<u>2.Configuration du Serveur OpenVPN:</u>

```
push "redirect-gateway def1 bypass-dhcp"
```

```
;push "dhcp-option DNS 208.67.222
push "dhcp-option DNS 8.8.8.8"
```

```
# TCP or UDP server?
proto tcp
;proto udp
```

```
# systems after creat
;user openvpn
;group openvpn
```

```
# configuration example.
ca ca.crt
cert server.crt
key server.key  # This file should be kept secret
```

Modifications du fichier server.conf pour forcer le client à utiliser le VPN pour son trafic et à passer par les serveurs DNS de google.

Utilisation du protocole TCP pour plus de fiabilité.

3. Génération des Clés et des Certificats pour le Serveur:

```
┌──(kali㉿kali)-[/etc/openvpn/easy-rsa]
└─$ sudo ./easyrsa build-ca nopass

Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
..+...+.......+......+.........................+..+.+.....
......+...+......+++++++++++++++++++++++++++++++++++++++*
......+..........+...+......+.....+....+......+...+......
.+..+...+.......+..+.+............+..+.+.................
++++++++++*.......+..+....+...+...+.....................
+...+....+...........+.......+..+...+............+.......
........+........+.......+......+..+..........+..+.+.....
.+...+.......+......+........+....+.....+......+.......+
.+...+................+.......+............+............
You are about to be asked to enter information that will
into your certificate request.
What you are about to enter is what is called a Distingui
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-R

Notice
------
CA creation complete. Your new CA certificate is at:
* /etc/openvpn/easy-rsa/pki/ca.crt

Build-ca completed successfully.
```

Génération du certificat CA

```
┌──(kali㉿kali)-[/etc/openvpn/easy-rsa]
└─$ sudo ./easyrsa build-server-full server nopass

Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
......++++++++++++++++++++++++++++++++++++++++*.+ ... +.
......+......+.....+....+.....+.+ ... + ... + ... +........
..+..+ ... + ... + ... +.+..................+......+..+....
... +.....+.......+.....+.+..+.+......+........+++++++
.....+.+..................+..+.......+..+....+......+
-----

Notice
------
Private-Key and Public-Certificate-Request files crea
Your files are:
* req: /etc/openvpn/easy-rsa/pki/reqs/server.req
* key: /etc/openvpn/easy-rsa/pki/private/server.key

You are about to sign the following certificate:

  Requested CN:     'server'
  Requested type:   'server'
  Valid for:        '825' days


subject=
    commonName                = server

Type the word 'yes' to continue, or any other input t
  Confirm requested details: yes

Using configuration from /etc/openvpn/easy-rsa/pki/9d
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'server'
Certificate is to be certified until Jul  5 18:13:45

Write out database with 1 new entries
Database updated

WARNING
=======
INCOMPLETE Inline file created:
* /etc/openvpn/easy-rsa/pki/inline/private/server.inl

Missing TLS Key, see 'easyrsa help gen-tls'


Notice
------
Certificate created at:
* /etc/openvpn/easy-rsa/pki/issued/server.crt
```

Génération du certificat du serveur

```
┌──(kali㉿kali)-[/etc/openvpn/easy-rsa]
└─$ sudo openssl dhparam -out dh.pem 2048

Generating DH parameters, 2048 bit long safe prime
.....................................................
.........................................+...........
.....................................................
.....................................................
.....................................................
.....................................................
.....................................................
.......+.............................................
.....................................................
.....................++++++++++++++++++++++++++++++
+++*
```

Génération des paramètres Diffie-Hellman.

4. Configuration du Client VPN:

```
naus@Akira:~$ ls -la
total 68
drwxr-x--- 9 naus naus 4096 Apr  2 20:51 .
drwxr-xr-x 3 root root 4096 Mar 26 09:03 ..
lrwxrwxrwx 1 naus naus   23 Mar 26 09:30 .aws -> /mnt/c/Users/na
lrwxrwxrwx 1 naus naus   25 Mar 26 09:30 .azure -> /mnt/c/Users/
-rw------- 1 naus naus 5239 Apr  2 17:09 .bash_history
-rw-r--r-- 1 naus naus  220 Mar 26 09:03 .bash_logout
-rw-r--r-- 1 naus naus 3771 Mar 26 09:03 .bashrc
drwx------ 3 naus naus 4096 Mar 26 12:49 .cache
drwxr-xr-x 5 naus naus 4096 Mar 26 12:57 .docker
drwxr-xr-x 3 naus naus 4096 Mar 26 12:39 .local
-rw-rw-r-- 1 naus naus    0 Apr  2 13:57 .motd_shown
-rw-r--r-- 1 naus naus  807 Mar 26 09:03 .profile
drwx------ 2 naus naus 4096 Apr  2 20:51 .ssh
-rw-r--r-- 1 naus naus    0 Mar 26 12:26 .sudo_as_admin_successf
-rw----r-x 1 naus naus 1208 Apr  2 20:51 ca.crt
-rw-r--r-- 1 naus naus  315 Apr  2 20:37 client.ovpn
-rw-r--r-- 1 naus naus  315 Apr  2 20:37 client1.conf
drwxr-xr-x 3 naus naus 4096 Mar 26 11:30 docker
drwxr-xr-x 8 naus naus 4096 Apr  1 21:05 docker-bench-security
drwx------ 4 naus naus 4096 Mar 26 12:49 snap
```

Configuration des fichiers sur le client VPN.

7. Vérification du Tunnel VPN:

```
naus@Akira:~$ ip a show tun0
3: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel
500
    link/none
    inet 10.8.0.2/24 brd 10.8.0.255 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::460b:a38e:b5b5:643e/64 scope link stable-privacy
       valid_lft forever preferred_lft forever
```

Nous pouvons ici observer que l'interface tun0 est bien apparue et que donc la connexion avec le serveur VPN fonctionne

```
naus@Akira:~$ ping 10.8.0.1
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data.
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=1.01 ms
64 bytes from 10.8.0.1: icmp_seq=2 ttl=64 time=1.58 ms
^C
--- 10.8.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.008/1.292/1.576/0.284 ms
naus@Akira:~$ traceroute 10.8.0.1
traceroute to 10.8.0.1 (10.8.0.1), 64 hops max
  1   10.8.0.1  0.894ms  0.755ms  0.680ms
naus@Akira:~$
```

Le tunnel VPN semble bien fonctionner.